University Department Information Management System (UDIIMS)

Use Case Document

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Use Case Index

ID	Title	Primary Actor	Module
UC-01	Student Authentication	Student	Module 1 — Academic
UC-02	Course Registration	Student	Module 1 — Academic
UC-03	GPA Calculation	Student	Module 1 — Academic
UC-04	Program & Backlog Tracking	Student	Module 1 — Academic
UC-05	Fee Status View	Student	Module 1 — Academic
UC-06	Faculty Management	Department Secretary	Module 2 — Department Portal
UC-07	Project Management	Department Secretary	Module 2 — Department Portal
UC-08	Inventory Management	Department Secretary	Module 2 — Department Portal
UC-09	Department Accounts	Department Secretary	Module 2 — Department Portal
UC-10	Course Offering Management	Department Secretary	Module 2 — Department Portal
UC-11	Grant Management	Finance Officer	Module 3 — Finance
UC-12	Consultancy Fund Management	Finance Officer	Module 3 — Finance
UC-13	Fee Collection & Modification	Finance Officer	Module 3 — Finance
UC-14	Project Financial Management	Finance Officer	Module 3 — Finance

Detailed Use Cases

Module 1: Academic Information System — UC-01: Student Authentication

ID:	UC-01
Title:	Student Authentication

Description:	The student provides their Roll Number to the system. The system retrieves and validates the student record from D1 Student Database and, if valid, grants dashboard access.
Primary Actor:	Student
Secondary Actor:	D1 Student Database
Pre-conditions:	1. The UDIIMS web application is accessible via a browser. 2. The student's Roll Number exists in D1 Student Database.
Post-conditions:	1. Student is authenticated and dashboard-access is set to true. 2. Student is granted access to the Student Dashboard (SGPA, Backlogs, Fee Status).
Dependency	Extended by: UC-02 (Course Registration), UC-03 (GPA Calculation), UC-04 (Program & Backlog Tracking), UC-05 (Fee Status View).
Generalization	None
Main Success Scenario:	1. Student enters their Roll Number (student-id) on the login screen. 2. System queries D1 Student Database using the student-id. 3. System retrieves student-record (student-id, student-name, program, semester, sgpa, cgpa, backlog-count, dashboard-access). 4. System sets dashboard-access = true. 5. System displays the Student Dashboard showing SGPA / CGPA / Backlogs / Fee Status.
Extensions / Alternate Flow:	1. 2a. Roll Number not found in D1: System displays 'Invalid Roll Number'. Access is denied. 2. 2b. D1 Student Database is unavailable: System displays 'Service Unavailable. Please try again later.' 3. 1a. Student submits an empty Roll Number field: System displays client-side validation error.
Frequency of Use:	Multiple times daily.
Status:	Approved
Owner:	Group 11
Priority:	High

Module 1: Academic Information System — UC-02: Course Registration

ID:	UC-02
Title:	Course Registration
Description:	The student submits a registration-request (student-id + semester-term + course codes). The system reads existing registrations from D2, updates registration records, and returns backlog status.
Primary Actor:	Student
Secondary Actor:	D2 Course Registration Database
Pre-conditions:	1. Student has been authenticated (UC-01) — dashboard-access = true. 2. The registration window for the current semester is open. 3. Course offerings for the semester exist in D2.

Post-conditions:	1. D2 Course Registration Database is updated (W: Registration Update). 2. Backlog Status is returned to the student.
Dependency	Includes: UC-01 (Student Authentication).
Generalization	None
Main Success Scenario:	1. Student submits registration-request = student-id + semester-term + {course-code}. 2. System reads existing registration records from D2 (R: Existing Registration). 3. System validates each course-code against available course offerings. 4. System writes the new/updated registration-record to D2 (W: Registration Update): registration-id, student-id, course-record, registration-status = active, backlog-flag. 5. System returns Backlog Status to the student.
Extensions / Alternate Flow:	1. 3a. Course-code does not exist in D2: System rejects that course and notifies the student. 2. 3b. Registration window has closed: System displays 'Registration period is over.' 3. 4a. Student attempts to register for a course they have already completed: System sets backlog-flag = false and skips re-registration. 4. 4b. Database write fails: System displays an error and rolls back the transaction.
Frequency of Use:	Once per semester during the registration period.
Status:	Approved
Owner:	Group 11
Priority:	High

Module 1: Academic Information System — UC-03: GPA Calculation

ID:	UC-03
Title:	GPA Calculation
Description:	Using grades and credits retrieved from D2 Course Registration Database, the system calculates SGPA and CGPA for the student and writes the updated GPA values back to D1 Student Database.
Primary Actor:	Student
Secondary Actor:	D2 Course Registration Database, D1 Student Database
Pre-conditions:	1. Student has been authenticated (UC-01). 2. Grades and credit data are available in D2 Course Registration Database (R: Grades & Credits).
Post-conditions:	1. SGPA and CGPA are computed and written to D1 Student Database (W: Updated GPA). 2. Updated SGPA / CGPA values are displayed on the student's dashboard.
Dependency	Includes: UC-01 (Student Authentication). Triggered by: UC-02 (Course Registration) via R: Grades & Credits.
Generalization	None

Main Success Scenario:	1. System retrieves R: Grades & Credits from D2 Course Registration Database for the student. 2. System calculates SGPA for the current semester: SGPA = Σ(Grade Points × Credits) / Σ(Credits). 3. System calculates CGPA as the cumulative weighted average across all completed semesters. 4. System writes W: Updated GPA (sgpa, cgpa, gpa-updated-timestamp) to D1 Student Database. 5. Updated SGPA / CGPA is returned to the Student Dashboard.
Extensions / Alternate Flow:	1. 1a. No grade data available in D2 for the student: System skips GPA calculation and displays 'Grades not yet available.' 2. 1b. Grade for a course is missing/null: System excludes that course from the SGPA calculation and flags it. 3. 4a. Write to D1 fails: System logs the error; existing GPA values remain unchanged.
Frequency of Use:	After every grade update or end-of-semester result declaration.
Status:	Approved
Owner:	Group 11
Priority:	High

Module 1: Academic Information System — UC-04: Program & Backlog Tracking

ID:	UC-04
Title:	Program & Backlog Tracking
Description:	The student queries the system using their Roll Number. The system reads course results from D2 Course Registration Database and displays completed courses, pending backlogs (grade = F), and a summary of credits earned versus required.
Primary Actor:	Student
Secondary Actor:	D2 Course Registration Database
Pre-conditions:	1. Student has been authenticated (UC-01). 2. Registration and grade records exist in D2 Course Registration Database.
Post-conditions:	1. Student can view a list of completed courses, pending backlogs, and credit summary.
Dependency	Includes: UC-01 (Student Authentication). Input from: UC-02 (Course Registration) via course results.
Generalization	None
Main Success Scenario:	1. Student submits Roll Number as query input. 2. System reads R: Course Results from D2 Course Registration Database. 3. System filters registration-records where registration-status = 'completed' — displays as completed courses. 4. System filters backlog-status: courses where grade = 'F' (backlog-flag = true) — displays as pending backlogs. 5. System computes total credits earned from completed courses versus total required credits.

	6. System displays the program progress summary to the student.
Extensions / Alternate Flow:	1. 2a. No records found in D2 for the student: System displays 'No academic records available.' 2. 4a. No backlogs found: System displays 'No pending backlogs.' in the backlog section.
Frequency of Use:	Multiple times per semester.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 1: Academic Information System — UC-05: Fee Status View

ID:	UC-05
Title:	Fee Status View
Description:	The system retrieves fee records from D5 Financial Database and displays the semester-wise fee status (fee amount, payment status, date) to the student. This is a read-only view.
Primary Actor:	Student
Secondary Actor:	D5 Financial Database
Pre-conditions:	1. Student has been authenticated (UC-01). 2. Fee records for the student exist in D5 Financial Database.
Post-conditions:	1. Student can view semester-wise fee details: amount, fee-status (paid/pending/partial/waived), and fee-updated-timestamp. 2. No modifications are possible from this view.
Dependency	Includes: UC-01 (Student Authentication). Fee data is written only by Finance Officer via UC-13 (Fee Collection & Modification).
Generalization	None
Main Success Scenario:	1. Student navigates to Fee Status from the dashboard. 2. System queries D5 Financial Database using student-id (R: Fee Status). 3. System retrieves fee-record = student-id + amount + fee-status + fee-updated-timestamp. 4. System displays semester-wise fee table to the student (Fee Details output). 5. Student reviews fee information — fields are read-only.
Extensions / Alternate Flow:	1. 2a. No fee records found for the student in D5: System displays 'No fee records available.' 2. 4a. Student attempts to edit fee data: System prevents any input — all fields are non-editable.
Frequency of Use:	At the start of each semester or when checking payment status.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 2: Department Portal — UC-06: Faculty Management

ID:	UC-06
Title:	Faculty Management
Description:	The Department Secretary sends Faculty Updates to the system. The system reads and writes faculty records to D3 Faculty Database and returns Faculty Records/Reports to the Secretary.
Primary Actor:	Department Secretary
Secondary Actor:	D3 Faculty Database
Pre-conditions:	1. Department Secretary is logged into the Department Portal. 2. Department context has been selected (dept_id is set).
Post-conditions:	1. Faculty record is added, updated, or deactivated in D3 Faculty Database (W: Faculty Record Update). 2. Faculty Records / Faculty Report is returned to the Department Secretary.
Dependency	Related to: UC-07 (Project Management) — faculty-id is used in project-details.
Generalization	None
Main Success Scenario:	1. Department Secretary sends Faculty Update (faculty-record data) to the system. 2. System reads existing R: Faculty Data from D3 Faculty Database. 3. For Add: Secretary enters faculty-record = faculty-id + faculty-name + designation + department-name + specialization + active-status. System writes to D3. 4. For Update: Secretary modifies existing faculty fields. System writes updated record to D3 (W: Faculty Record Update). 5. For Remove: System sets active-status = false for the selected faculty record in D3. 6. System returns Faculty Records / Faculty Report to the Department Secretary.
Extensions / Alternate Flow:	1. 3a. Duplicate faculty-id on Add: System displays 'Faculty ID already exists.' 2. 3b. Mandatory fields are empty: System displays validation error. 3. 5a. Faculty has active project associations on Remove: System warns 'Faculty has active projects. Confirm removal?'
Frequency of Use:	Occasionally — when faculty join, leave, or change designation.
Status:	Approved
Owner:	Group 11
Priority:	High

Module 2: Department Portal — UC-07: Project Management

ID:	UC-07
Title:	Project Management

Description:	The Department Secretary provides Project Details to the system. The system reads Faculty Info from D3 Faculty Database (for supervisor assignment) and writes Project Records to D3. Project Reports are returned to the Secretary.
Primary Actor:	Department Secretary
Secondary Actor:	D3 Faculty Database
Pre-conditions:	1. Department Secretary is logged into the Department Portal. 2. Faculty records exist in D3 Faculty Database for supervisor assignment.
Post-conditions:	1. Project record is added or updated in D3 (W: Project Records). 2. Project Report is returned to the Department Secretary.
Dependency	Reads: R: Faculty Info from D3 (for supervisor validation). Related to: UC-06 (Faculty Management).
Generalization	None
Main Success Scenario:	1. Department Secretary inputs Project Details to the system. 2. System reads R: Faculty Info from D3 Faculty Database to validate the supervising faculty-id. 3. System creates or updates project-details = project-id + project-title + faculty-id + project-budget + project-status. 4. System writes W: Project Records to D3 Faculty Database. 5. System returns Project Report to the Department Secretary. 6. Secretary can click a publication link associated with the project; system opens it in the default browser.
Extensions / Alternate Flow:	1. 2a. Supervising faculty-id not found in D3: System displays 'Invalid Faculty ID.' 2. 3a. Duplicate project-id: System displays 'Project ID already exists.' 3. 4a. Write to D3 fails: System displays error and does not save the project record.
Frequency of Use:	Periodically — when new projects are initiated or existing ones updated.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 2: Department Portal — UC-08: Inventory Management

ID:	UC-08
Title:	Inventory Management
Description:	The Department Secretary sends Inventory Updates to the system. The system reads and writes inventory records to D4 Inventory Database and returns Inventory Reports to the Secretary.
Primary Actor:	Department Secretary
Secondary Actor:	D4 Inventory Database
Pre-conditions:	1. Department Secretary is logged into the Department Portal. 2. Department context has been selected.

	3. For lab inventory: applicable only to science/technical departments (e.g., CSE, Physics).
Post-conditions:	1. Inventory record is added, updated, or disposed in D4 Inventory Database (W: Inventory Update). 2. Inventory Report is returned to the Department Secretary.
Dependency	None
Generalization	None
Main Success Scenario:	1. Department Secretary sends Inventory Update to the system. 2. System reads existing R: Inventory Data from D4 Inventory Database. 3. For Add: Secretary enters inventory-record = item-id + item-name + category + quantity + location + condition + assigned-department. System writes to D4. 4. For Update: Secretary modifies existing inventory fields (e.g., condition from 'good' to 'fair'). System writes updated record to D4. 5. For Remove: System sets condition = 'disposed' for the item in D4. 6. System returns Inventory Report to the Department Secretary.
Extensions / Alternate Flow:	1. 3a. Secretary from a non-technical department attempts lab inventory entry: System displays 'Lab Inventory not applicable for this department.' 2. 3b. Duplicate item-id: System displays 'Item ID already exists.' 3. 3c. Required fields are empty: System displays validation error.
Frequency of Use:	Monthly or when assets are acquired, transferred, or disposed.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 2: Department Portal — UC-09: Department Accounts

ID:	UC-09
Title:	Department Accounts
Description:	The system reads Budget/Project Finance Data from D5 Financial Database and generates a Department Financial Report for the Department Secretary. This is a read-only view of financial data.
Primary Actor:	Department Secretary
Secondary Actor:	D5 Financial Database
Pre-conditions:	1. Department Secretary is logged into the Department Portal. 2. Financial records for the department exist in D5 Financial Database.
Post-conditions:	1. Department Financial Report is displayed to the Secretary: budget-allocation, total-expenses, remaining-balance. 2. If remaining balance is negative, a warning is shown.
Dependency	Reads: R: Budget / Project Finance Data from D5. Financial data is written only by Finance Officer (UC-10, UC-11, UC-13).
Generalization	None

Main Success Scenario:	1. Department Secretary navigates to Department Accounts. 2. System reads R: Budget / Project Finance Data from D5 Financial Database, filtered by dept-id. 3. System retrieves department-financial-report = department-id + budget-allocation + total-expenses + remaining-balance. 4. System verifies: Funds Remaining = (Opening + Grants + Income) − Expenses. 5. System displays Department Financial Report to the Secretary (read-only).
Extensions / Alternate Flow:	1. 2a. No financial records for department in D5: System displays 'No financial data available for this department.' 2. 4a. Computed remaining-balance is negative: System displays warning 'Warning: Funds Remaining is negative. Budget review required.'
Frequency of Use:	Monthly or during budget review periods.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 2: Department Portal — UC-10: Course Offering Management

ID:	UC-10
Title:	Course Offering Management
Description:	The Department Secretary creates or updates Course Offerings. The system reads existing course data from D2 Course Registration Database, updates it, and returns a Course Offering Report.
Primary Actor:	Department Secretary
Secondary Actor:	D2 Course Registration Database
Pre-conditions:	1. Department Secretary is logged into the Department Portal. 2. D2 Course Registration Database is accessible.
Post-conditions:	1. Course offering is added or updated in D2 (W: Course Offering Update). 2. Course Offering Report is returned to the Department Secretary. 3. Updated course offerings become available for student registration (UC-02).
Dependency	Writes to D2 which is read by UC-02 (Course Registration).
Generalization	None
Main Success Scenario:	1. Department Secretary sends Create/Update Course Offering request. 2. System reads R: Existing Course Data from D2 Course Registration Database. 3. Secretary enters course-offering-update = course-code + course-name + credit-hours + semester-term. 4. System validates input and writes W: Course Offering Update to D2. 5. System returns Course Offering Report to the Department Secretary.
Extensions / Alternate Flow:	1. 3a. Duplicate course-code for the same semester-term: System displays 'Course already offered this term.' 2. 3b. Invalid credit-hours (zero or negative): System displays validation error.

	3. 4a. Removing a course that has enrolled students: System warns 'Students enrolled in this course. Confirm removal?'
Frequency of Use:	Once per semester during academic planning.
Status:	Approved
Owner:	Group 11
Priority:	High

Module 3: Finance Section — UC-11: Grant Management

ID:	UC-11
Title:	Grant Management
Description:	The Finance Officer enters Grant Entry data. The system writes Grant Updates to D5 Financial Database and reads Existing Grant Records to return a Grant Report to the Finance Officer.
Primary Actor:	Finance Officer
Secondary Actor:	D5 Financial Database
Pre-conditions:	1. Finance Officer is logged into the Finance Section. 2. D5 Financial Database is accessible.
Post-conditions:	1. grant-entry is recorded in D5 as a financial-record with record-type = 'grant' (W: Grant Update). 2. Department budget allocation is updated. 3. Grant Report is returned to the Finance Officer.
Dependency	Writes to D5 which is read by UC-09 (Department Accounts).
Generalization	None
Main Success Scenario:	1. Finance Officer submits Grant Entry to the system. 2. System reads R: Existing Grant Records from D5 Financial Database. 3. System validates the grant-entry = grant-id + amount + transaction-date + department-id. 4. System writes W: Grant Update to D5 as financial-record (record-type = 'grant'). 5. System updates the department's budget-allocation in D5. 6. System returns Grant Report (financial-report) to the Finance Officer.
Extensions / Alternate Flow:	1. 3a. Duplicate grant-id: System displays 'Grant ID already exists.' 2. 3b. Invalid department-id: System displays 'Department not found.' 3. 4a. Write to D5 fails: System displays error and does not save the record.
Frequency of Use:	Whenever a new grant is received (irregular, project-dependent).
Status:	Approved
Owner:	Group 11
Priority:	High

Module 3: Finance Section — UC-12: Consultancy Fund Management

ID:	UC-12
Title:	Consultancy Fund Management
Description:	The Finance Officer enters Consultancy Entry data. The system writes Consultancy Updates to D5 Financial Database and returns a Consultancy Report. Department-wise consultancy revenue is tracked.
Primary Actor:	Finance Officer
Secondary Actor:	D5 Financial Database
Pre-conditions:	1. Finance Officer is logged into the Finance Section. 2. D5 Financial Database is accessible.
Post-conditions:	1. consultancy-entry is recorded in D5 as financial-record with record-type = 'consultancy' (W: Consultancy Update). 2. Consultancy Report is returned to the Finance Officer.
Dependency	Writes to D5. Related to UC-09 (Department Accounts) — consultancy income reflected in department balance.
Generalization	None
Main Success Scenario:	1. Finance Officer submits Consultancy Entry to the system. 2. System reads R: Consultancy Records from D5 Financial Database. 3. System validates consultancy-entry = consultancy-id + amount + transaction-date + department-id. 4. System writes W: Consultancy Update to D5 as financial-record (record-type = 'consultancy'). 5. System returns Consultancy Report to the Finance Officer.
Extensions / Alternate Flow:	1. 3a. Duplicate consultancy-id: System displays 'Consultancy ID already exists.' 2. 3b. Amount is zero or negative: System displays 'Amount must be a positive value.' 3. 4a. Write to D5 fails: System displays error and rolls back.
Frequency of Use:	As needed when consultancy income is received.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Module 3: Finance Section — UC-13: Fee Collection & Modification

ID:	UC-13
Title:	Fee Collection & Modification
Description:	The Finance Officer sends a Fee Status Update. The system reads the Student Fee Record from D5 Financial Database, updates the fee status, and writes the Updated Fee Status back to D5. Fee Confirmation is sent to the Finance Officer.
Primary Actor:	Finance Officer
Secondary Actor:	D5 Financial Database

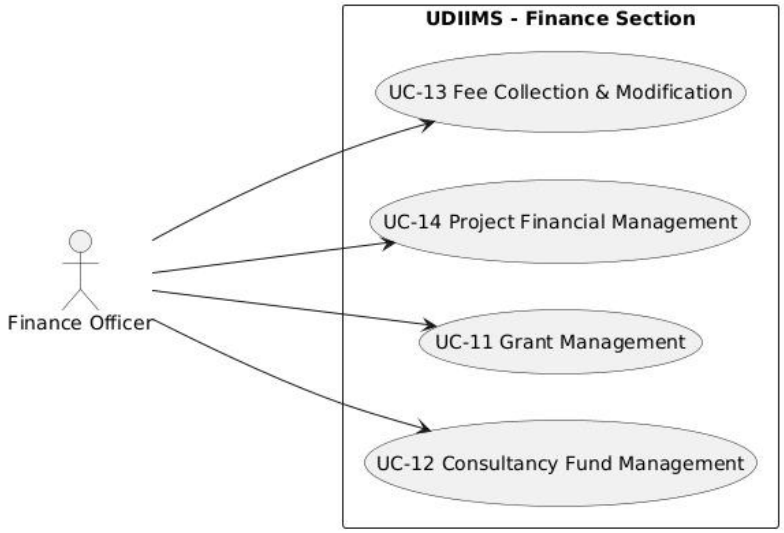
Pre-conditions:	1. Finance Officer is logged into the Finance Section. 2. Student Roll Number is valid. 3. Fee record for the student exists in D5 Financial Database.
Post-conditions:	1. fee-record in D5 is updated with new fee-status and fee-updated-timestamp (W: Updated Fee Status). 2. Fee Confirmation is returned to the Finance Officer. 3. Student's Fee Status View (UC-05) immediately reflects the updated status.
Dependency	Writes to D5 which is read by UC-05 (Fee Status View — Student).
Generalization	None
Main Success Scenario:	1. Finance Officer submits Fee Status Update with Student Roll Number. 2. System reads R: Student Fee Record from D5 Financial Database. 3. System retrieves fee-record = student-id + amount + fee-status + fee-updated-timestamp. 4. Finance Officer selects 'Mark as Paid' or 'Mark as Due'. 5. System writes W: Updated Fee Status to D5 — updates fee-status and fee-updated-timestamp. 6. System returns Fee Confirmation to the Finance Officer. 7. Student's Fee Status View is immediately updated.
Extensions / Alternate Flow:	1. 1a. Invalid Roll Number — student not found in D5: System displays 'Student record not found.' 2. 2a. No fee record found for the student: System displays 'No fee record found.' 3. 5a. Write to D5 fails: System displays error; previous fee-status is preserved. 4. Note: Department Secretary has read-only access to fee data. Only Finance Officer can modify.
Frequency of Use:	At the start of each semester and when fee payments are processed.
Status:	Approved
Owner:	Group 11
Priority:	High

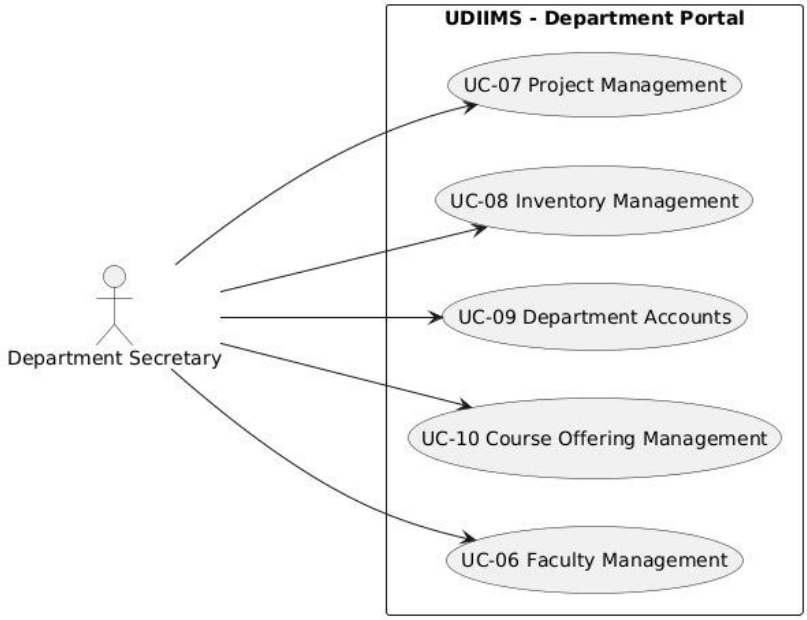
Module 3: Finance Section — UC-14: Project Financial Management

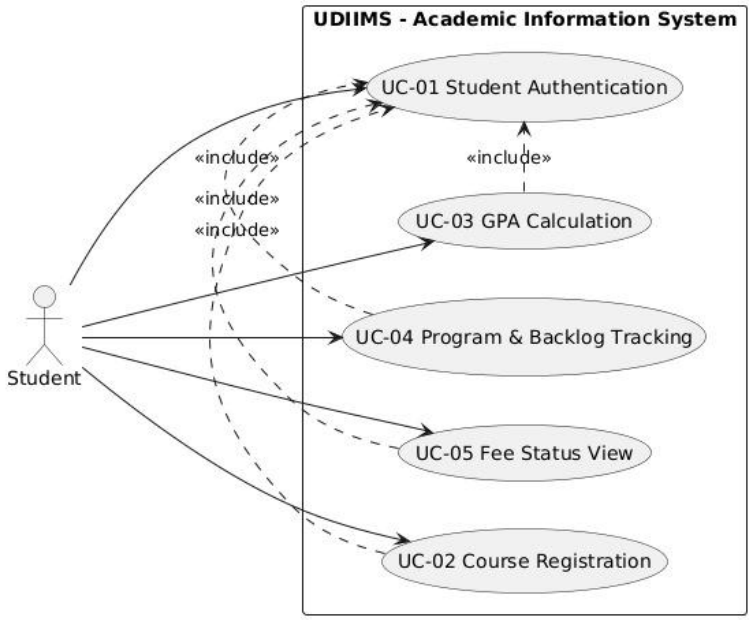
ID:	UC-14
Title:	Project Financial Management
Description:	The Finance Officer enters Project Budget Allocation / Expense data. The system reads and writes Project Finance Records to D5 Financial Database and returns a Project Financial Report.
Primary Actor:	Finance Officer
Secondary Actor:	D5 Financial Database
Pre-conditions:	1. Finance Officer is logged into the Finance Section. 2. Project records exist in the system with valid project-ids. 3. D5 Financial Database is accessible.
Post-conditions:	1. project-financial-entry is recorded in D5 (W: Project Finance Update). 2. Project Financial Report is returned to the Finance Officer.

	3. If project budget is exceeded, a warning is issued.
Dependency	Writes to D5. Related to UC-09 (Department Accounts) — project expenses affect department balance.
Generalization	None
Main Success Scenario:	1. Finance Officer submits Project Budget Allocation / Expense Entry. 2. System reads R: Project Finance Records from D5 Financial Database. 3. System validates project-financial-entry = project-id + project-budget + expense-amount + transaction-date. 4. System writes W: Project Finance Update to D5 as financial-record (record-type = 'project-budget' or 'expense'). 5. System verifies: Funds Remaining must not go negative. If it does, a warning is shown. 6. System returns Project Financial Report to the Finance Officer.
Extensions / Alternate Flow:	1. 3a. Invalid project-id: System displays 'Project not found.' 2. 5a. Expense causes negative project balance: System displays 'Warning: Project budget will be exhausted. Confirm entry?' 3. 4a. Write to D5 fails: System displays error and rolls back.
Frequency of Use:	Monthly or when project expenses are incurred.
Status:	Approved
Owner:	Group 11
Priority:	Medium

Use Case Diagram-


UDIIMS - Finance Section
UC-13 Fee Collection & Modification
UC-14 Project Financial Management
UC-11 Grant Management
UC-12 Consultancy Fund Management
Finance Officer


UDIIMS - Department Portal
UC-07 Project Management
UC-08 Inventory Management
UC-09 Department Accounts
UC-10 Course Offering Management
UC-06 Faculty Management
Department Secretary


UDIIMS - Academic Information System
UC-01 Student Authentication
«include»
«include»
«include»
«include»
UC-03 GPA Calculation
UC-04 Program & Backlog Tracking
UC-05 Fee Status View
UC-02 Course Registration
Student